Markscheme

Mathematics: analysis and approaches
Practice question bank

72. (a)

A geometric sequence has $u_3 = 20$ and $u_6 = 160$.

Find the common ratio, the first term, and the least value of n for which $u_n > 100\,000$.

$$\frac{u_6}{u_3} = r^3 = \frac{160}{20} = 8 \Rightarrow r = 2$$ *(M1)* **A1**

$$u_1 = \frac{u_3}{r^2} = \frac{20}{4} = 5$$ **A1**

solve $5(2)^{n-1} > 100\,000$: $2^{n-1} > 20\,000$ *(M1)*

$$n - 1 > \log_2(20\,000) = 14.29\ldots$$ *(M1)* **A1**

testing integers confirms $n = 16$ **A1**

[7 marks]